



CS111L – Programming Fundamentals

LAB # 02

Arithmetic Expression

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Task 1: Parentheses Precedence

A program to evaluate the following:

1. $5 + 2 * 3$
2. $(5 + 2) * 3$
3. $20 / (2 + 3)$

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     cout << 5 + 2 * 3 << endl;
6     cout << (5 + 2) * 3 << endl;
7     cout << 20 / (2 + 3);
8
9     return 0;
10 }
```

Output

11
21
4

Task 2: Unary Plus/Minus

A number and:

1. Its positive value using +n
2. Its negated value using -n

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n;
6
7     cout << "Enter a number: ";
8     cin >> n;
9
10    cout << "Positive: " << +n << endl;
11    cout << "Negated: " << -n;
12
13    return 0;
14
15 }
```

Output

```
Enter a number: 13
Positive value: 13
Negated value: -13
```

Task 3: Multiplication and Division

Two numbers and calculation:

1. Product $a * b$
2. Quotient a / b (integer division)
3. Division result as floating point (float) a / b

```
1 #include <iostream>
2 using namespace std;
3
4 int main(){
5     int a, b;
6
7     cout << "Enter two numbers: ";
8     cin >> a >> b;
9
10    cout << "Product: " << a * b << endl;
11    cout << "Quotient: " << a / b << endl;
12    cout << "Floating Point: " << (float)a / b;
13
14    return 0;
15 }
```

Output

```
Enter two numbers: 13
45
Product: 585
Quotient: 0
Floating Point: 0.288889
```

Task 4: Modulus Practice

Two integers x and y. Display:

1. $x \% y$
2. $(x + y) \% 5$
3. $(x * y) \% 10$

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int x, y;
6
7     cout << "Enter two integers: ";
8     cin >> x >> y;
9
10    cout << "x % y = " << x % y << endl;
11    cout << "(x + y) % 5 = " << (x + y) % 5 << endl;
12    cout << "(x * y) % 10 = " << (x * y) % 10;
13
14    return 0;
15 }
```

Output

```
Enter two integers: 13
28
x % y = 13
(x + y) % 5 = 1
(x * y) % 10 = 4
```

Task 5: Expression Evaluation

Evaluation and the result of each:

1. $10 + 20 * 3 - 5$
2. $(10 + 20) * (3 - 5)$
3. $100 / 10 \% 3 + 4 * 2$

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     cout << 10 + 20 * 3 - 5 << endl;
6     cout << (10 + 20) * (3 - 5) << endl;
7     cout << 100 / 10 % 3 + 4 * 2;
8     return 0;
9 }
```

Output

65
-60
9

Task 6: Complex Expression

Three numbers a, b, c:

- $\text{result} = (a * b) + (c \% a) - (-b)$

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int a, b, c, result;
6
7     cout << "Enter three numbers: ";
8     cin >> a >> b >> c;
9
10    result = (a * b) + (c % a) - (-b);
11
12    cout << "Result = " << result;
13
14    return 0;
15 }
```

Output

```
Enter three numbers: 21
24
43
Result = 529
```

Task 7: Salary Calculation (Gross & Net Pay)

Input:

- Basic Pay
- Travel Allowance
- House Rent
- Daily Allowance
- Tax percentage

Calculate:

- Gross Pay = Basic Pay + Travel Allowance + House Rent + Daily Allowance
- Net Pay = Gross Pay - (Gross Pay * Tax% / 100)
- Output Net Pay.

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     float basicPay, travelAllowance, houseRent;
6     float dailyAllowance, tax, grossPay, netPay;
7
8     cout << "Enter Basic Pay: ";
9     cin >> basicPay;
10
11     cout << "Enter Travel Allowance: ";
12     cin >> travelAllowance;
13
14     cout << "Enter House Rent: ";
15     cin >> houseRent;
16
17     cout << "Enter Daily Allowance: ";
18     cin >> dailyAllowance;
19
20     cout << "Enter Tax Percentage: ";
21     cin >> tax;
22
23     grossPay = basicPay + travelAllowance + houseRent + dailyAllowance;
24     netPay = grossPay - (grossPay * tax / 100);
25
26     cout << "Gross Pay = " << grossPay << endl;
27     cout << "Net Pay = " << netPay;
28     return 0;
29 }
```

Output

```
Enter Basic Pay: 100000
Enter Travel Allowance: 25000
Enter House Rent: 25000
Enter Daily Allowance: 1200
Enter Tax Percentage: 99
Gross Pay = 151200
Net Pay = 1512
```

Task 8: Student Marks Percentage

Marks of 5 subjects:

- Total Marks = sum of 5 subjects
- $\text{Percentage} = (\text{Total Marks} / 500.0) * 100$
- Output percentage.

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     float marks1, marks2, marks3, marks4, marks5;
6     float total, percentage;
7
8     cout << "Enter marks of 5 subjects: ";
9     cin >> marks1 >> marks2 >> marks3 >> marks4 >> marks5;
10
11     total = marks1 + marks2 + marks3 + marks4 + marks5;
12     percentage = (total / 500.0) * 100;
13
14     cout << "Total Marks = " << total << endl;
15     cout << "Percentage = " << percentage << "%";
16     return 0;
17 }
```

Output

```
Enter marks of 5 subjects: 99
99
98
97
12
Total Marks = 405
Percentage = 81%
```

Task 9: Shop Bill with Discount

Input:

- Price of an item
- Quantity
- Discount percentage

Calculate:

- $\text{Bill} = \text{Price} * \text{Quantity}$
- $\text{Discount} = (\text{Bill} * \text{Discount}\%) / 100$
- $\text{Net Bill} = \text{Bill} - \text{Discount}$
- Output Net Bill.

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     float price, quantity, discountPercentage;
6     float bill, discount, netBill;
7
8     cout << "Enter Price of Item: ";
9     cin >> price;
10
11     cout << "Enter Quantity: ";
12     cin >> quantity;
13
14     cout << "Enter Discount Percentage: ";
15     cin >> discountPercentage;
16
17     bill = price * quantity;
18     discount = (bill * discountPercentage) / 100;
19     netBill = bill - discount;
20
21     cout << "Bill = " << bill << endl;
22     cout << "Discount = " << discount << endl;
23     cout << "Net Bill = " << netBill;
24
25     return 0;
26 }
```

Output

```
Enter Price of Item: 1000
Enter Quantity: 5
Enter Discount Percentage: 13
Bill = 5000
Discount = 650
Net Bill = 4350
```

Task 10: Electricity Bill

Input:

- Units consumed
- Price per unit
- Tax percentage

Calculate:

- $\text{Bill} = \text{Units} * \text{Price per unit}$
- $\text{Net Bill} = \text{Bill} + (\text{Bill} * \text{Tax\%} / 100)$

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     float units, pricePerUnit, taxPercentage;
6
7     cout << "Enter Units Consumed: ";
8     cin >> units;
9
10    cout << "Enter Price Per Unit: ";
11    cin >> pricePerUnit;
12
13    cout << "Enter Tax Percentage: ";
14    cin >> taxPercentage;
15
16    float bill, netBill;
17
18    bill = units * pricePerUnit;
19    netBill = bill + (bill * taxPercentage / 100);
20
21    cout << "Bill = " << bill << endl;
22    cout << "Net Bill = " << netBill;
23
24    return 0;
25 }
```

Output

```
Enter Units Consumed: 2
Enter Price Per Unit: 999
Enter Tax Percentage: 9999
Bill = 1998
Net Bill = 201778
```